Product Design

Graph Generation with Sonify

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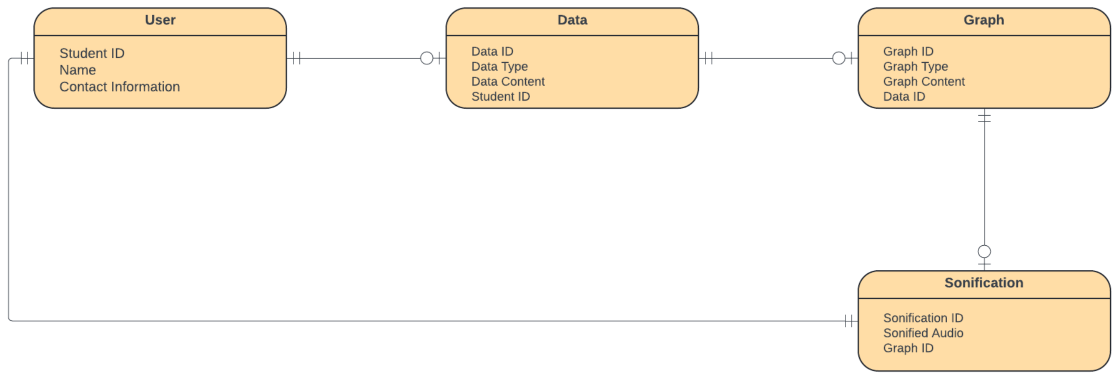
Revision Number	*Revision Date*	*Summary of Changes*	*Author(s)*
0.1	**09/26/2024**	**Initial description of classes added in class diagram.**	**Dillon Edington**
0.2	**10/6/2024**	**Initial sitemap added to the Information Architecture section**	**John Whitgrove**
0.3	**10/06/2024**	**Version 1 Wireframe Design**	**Katie Williams**
0.4	**10/06/2024**	**Sequence diagrams of each use case.**	**Isaac Rea**
0.5	**10/06/2024**	**Design Summary Added**	**Dillon Wright**
0.6	**10/06/2024**	**ER Diagram Added**	**Sain Madrigal**
1.0	**11/21/2024**	**Updated New Wireframe Screenshots, Added Design Rationale for New Design**	**Katie Williams**
1.1	**11/23/2024**	**Updated Sequence Diagram**	**Isaac Rea**
1.2	**12/8/2024**	**Updated Select and View Graph Screenshots. Update Design Rationale.**	**Katie Williams**
2.0	**2/8/2025**	**Updated all Screenshots and Design Rationale**	**Katie Williams**
2.1	**2/28/2025**	**Updated Screenshots and Design Rationale**	**Katie Williams**
2.2	**3/14/2025**	**Updated Screenshots and Design Rationale**	**Katie Williams**

Class Diagram(s)

- **Data –** This class contains the data entered by the user in a 2D array, as well as other elements suited for storing a spreadsheet, such as column names and data size.
- **Graph –** This class will be associated with one or more Data classes and will have functions to display one or more of the associated Data classes as conventional graph types. There will be functions that verify that Data is capable of being displayed as given graph types, and if so Graph will format the data to this end.
- **BarGraph** – This class will be a child of Graph and contain elements and functions unique to the representation of a bar graph, such as axisScale and barColor.
- **PieGraph** - This class will be a child of Graph and contain elements and functions unique to the representation of a pie graph, such as pieRadius and sliceColor.

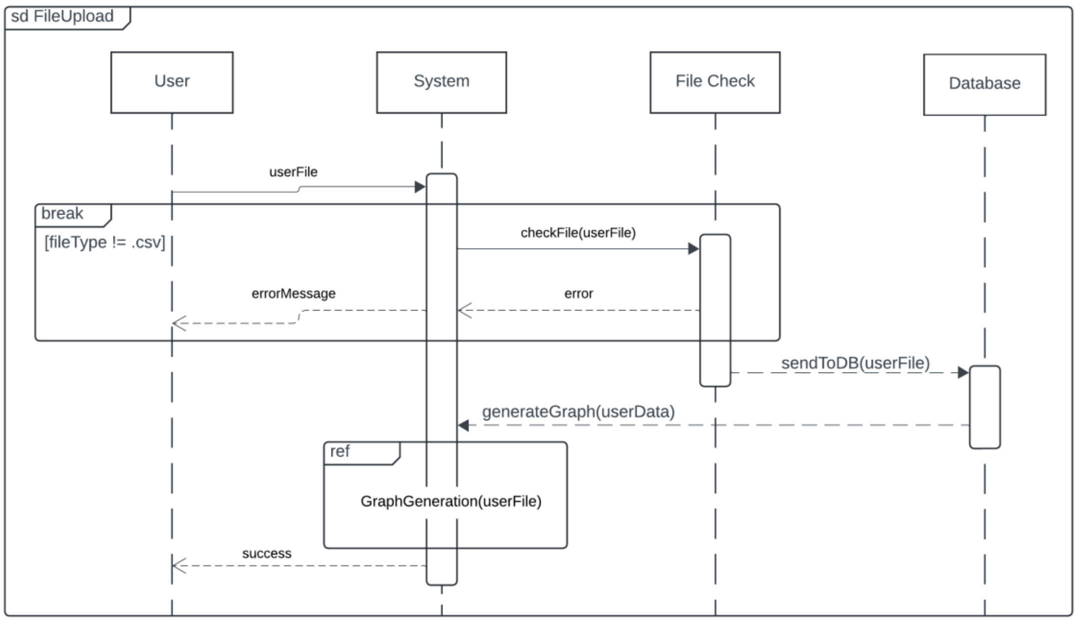
- **ScatterGraph –** This class will be a child of Graph and contain elements and functions unique to the representation of a scatter plot, such as axisScale and densityScale.
- **Sound –** Each Sound class will be associated with one Graph class and will be responsible for the sonification of each Data represented in the Graph. Will contain elements such as lowestPitch, highestPitch, key, and other musically significant qualities.

ER Diagram(s)

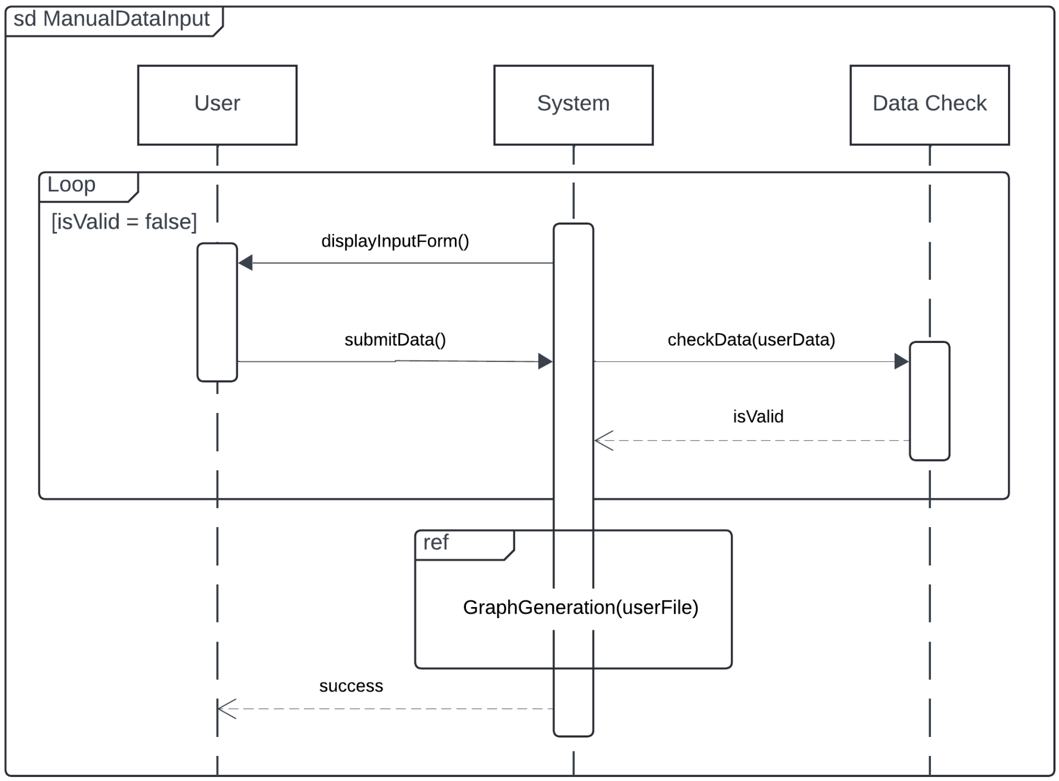


Sequence Diagrams

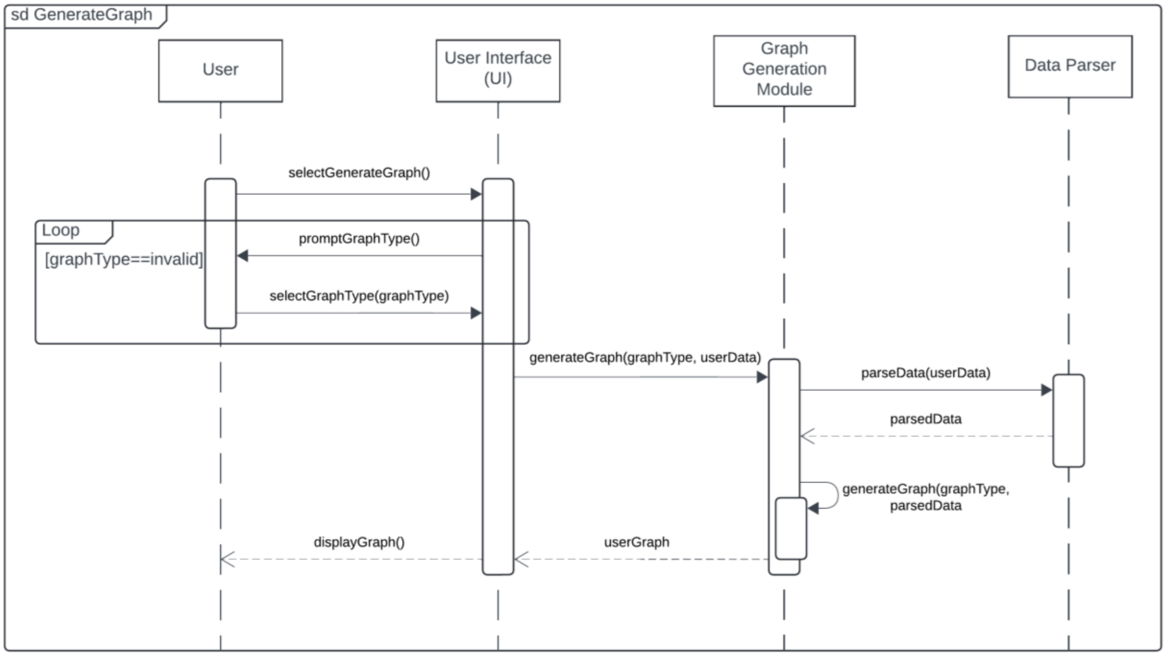
1. **Upload Data: UC-01**



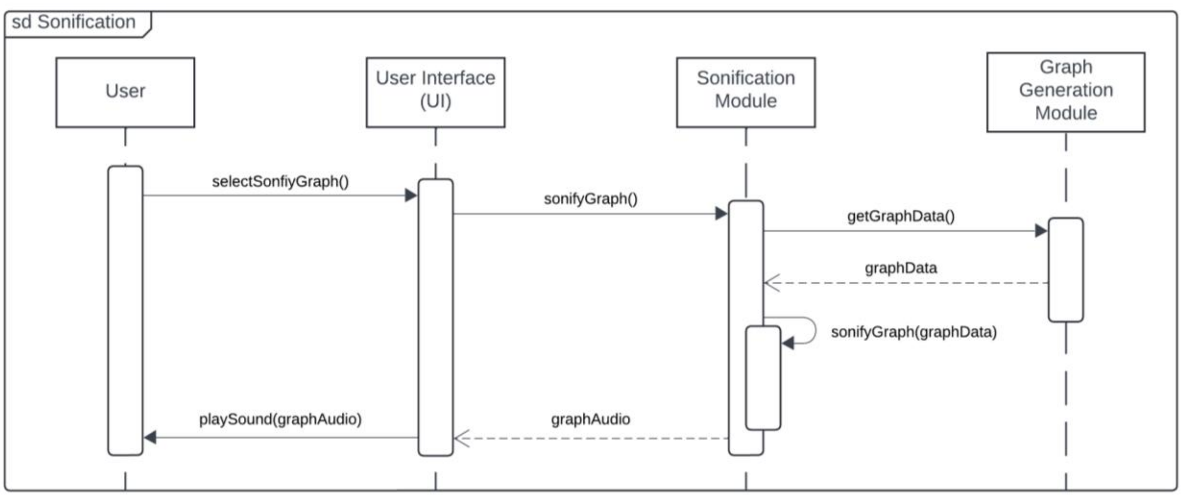
2. Manual Data Input: UC-02



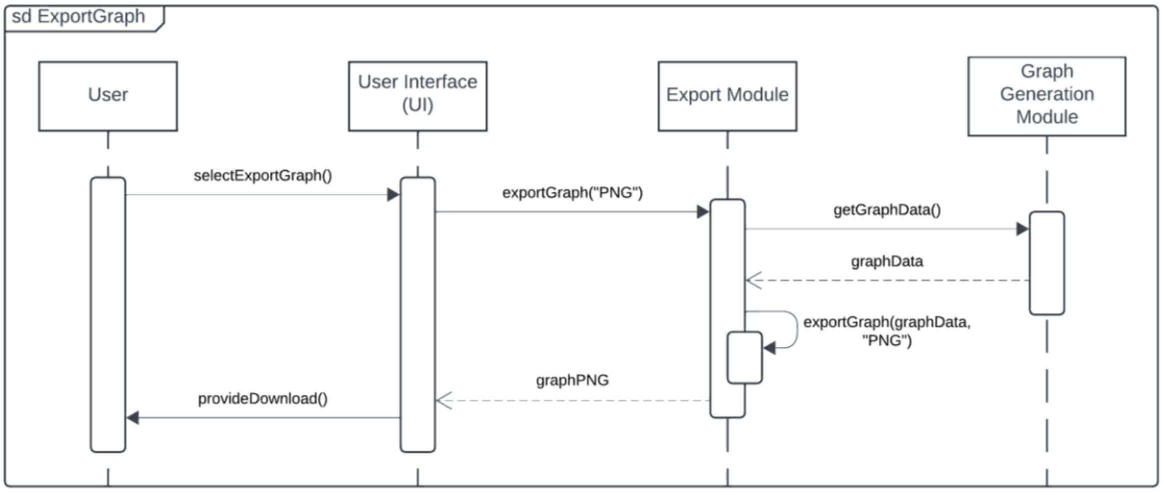
3. Generate Graph: UC-03



4. Sonification: UC-04

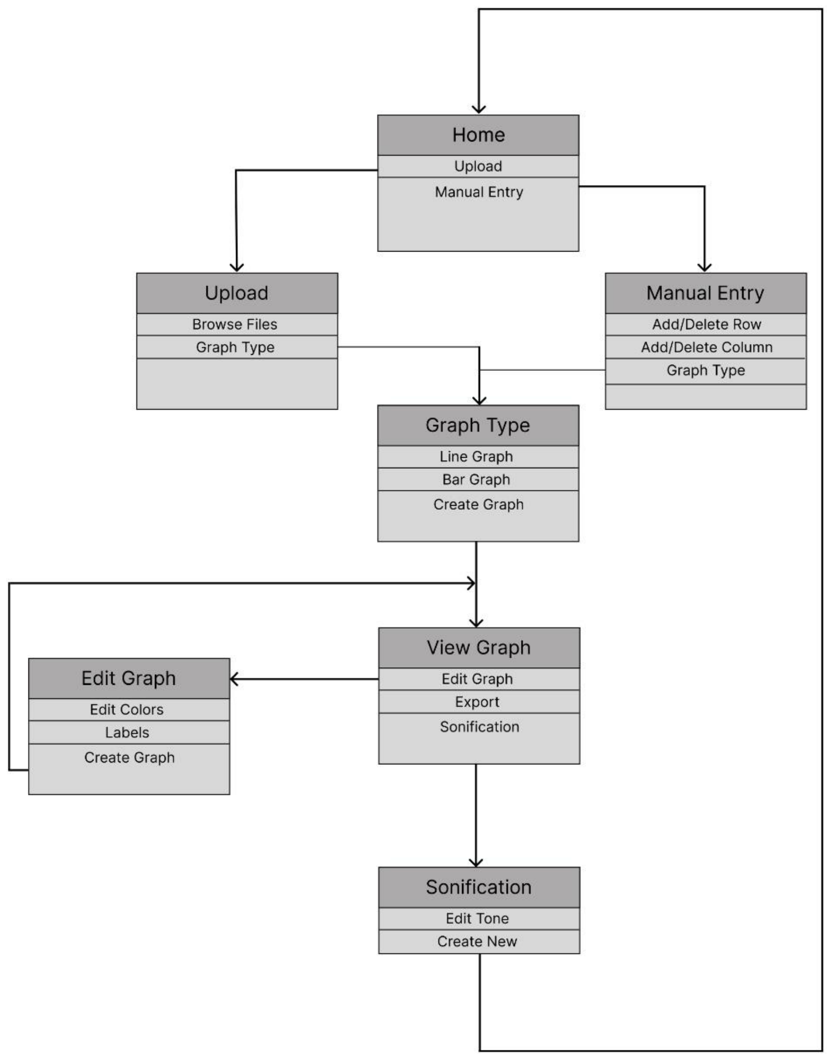


5. Export Generated Graph: UC-05


sd ExportGraph
User
User Interface (UI)
Export Module
Graph Generation Module
selectExportGraph()
exportGraph("PNG")
getGraphData()
graphData
exportGraph(graphData, "PNG")
graphPNG
provideDownload()

Information Architecture Diagram

Sitemap

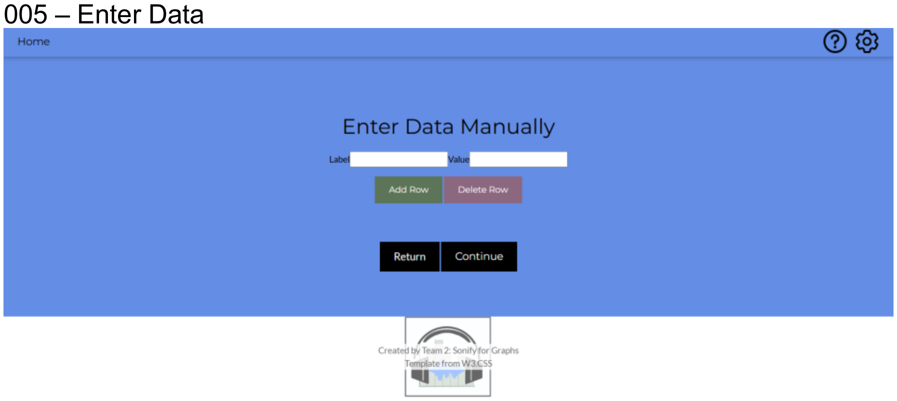


User Interface Wireframe Screenshot

001 - Homepage

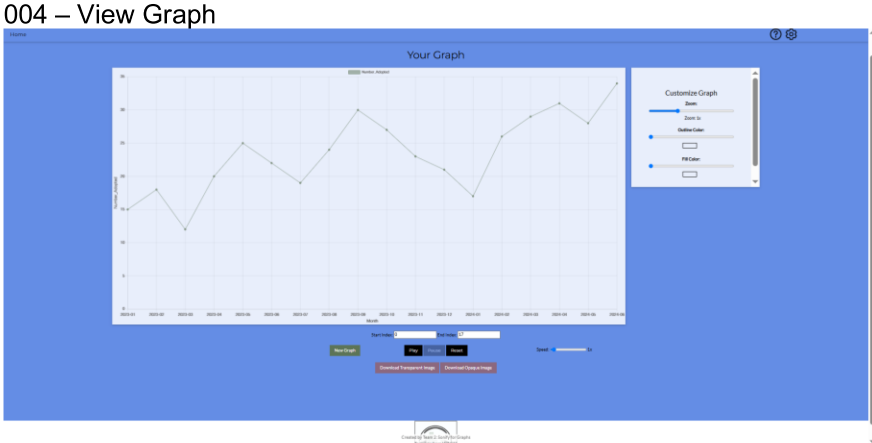
002 - Upload

005 – Enter Data



003 – Graph Selection

004 – View Graph



Design Summary

The website will start with a home screen that allows for either a manual graph creation entry, or an upload of a preexisting graph. Through the upload pathway, the user will be able to browse files for graph selection or an option to select a graph type to upload. If a user decides to manually enter a graph, rows and columns may be added/removed along with the selection of a graph type.

The graph type will be created upon user choice, which will then be ready for viewing/editing. The user can then edit, label, create another graph, or send it through the process of sonification. Then, the graph can be further edited through tone or exported as a .pdf file. Finally, after exporting the graph, the website allows you to create/upload a new graph if needed.

The main dynamic of the website will be the User/U.I. relationship loop of graph creation, as well as the sonification process.

Design Rationale

Rationale for Update 2.2

Some small features have been added to the overall design of the project. The first thing you notice is that a project logo has been added to the footer of each web page. This is to give the project a more professional look as most companies include their logos in their website. The graph selection UI has also been changed slightly. Instead of just words, icons for each graph type have been included. This is to make the graph selection process easier so that users can see examples of what type of graph they are choosing.

Rationale for Update 2.1

The look of the product has stayed relatively the same, although the color scheme has been changed. It has been changed to a more pleasing blue with some green and purple included with some buttons. Black and White has remained included as well. The Bootstrap framework has also been included with the inputs used for the graph type selection. Instead of checkboxes, they are now toggleable buttons. They are still exclusive, so only one graph type can be selected. This is to increase the overall look of the web application.

Rationale for Update 2.0

The look of the product has changed drastically from the 1.0 versions. The sponsor requested that the web application be given a more contemporary look. That is the main reason why the visual design of the UI has changed. A new template from W3 has been introduced. However, the flow of the app has remained the same. The plan is to continue changing components to make the app as visually appealing as possible.

Rationale for Update 1.0

Overall, the design has stayed relatively the same between the 0.6 and 1.0 versions. The main difference is the addition of the top bar that will hold the project logo, when designed, and information icon, and a settings icon. The purpose of the top bar is to be more consistent with other websites on the Internet. Will research design templates to give the website a more contemporary look,

A few new interaction points were added to enhance the graph creation and sonification. One was a user text input that takes the user’s name of the graph and adds it to the graph creation. The other buttons are on the view graph page that pauses the audio and the other resets the audio to play from the beginning.